

1. ^{ارزیابی} Assessment of the patient's condition

1 What methods of general examination are used to assess the patient's condition? (1,2,3,4,5)

1. Assessment of the patient's ^{هوشیاری} consciousness; 2. Assessment of the patient's position; 3. Evaluation of facial expressions; 4. Assessment of the condition of the skin and visible mucous membranes; 5. Determination of body temperature.

2 List the options for the patient's condition. (1,2,3,4)

1. Satisfactory; 2. Moderate; 3. Heavy; 4. Extremely heavy.

3 ^{رضایت بخش} For what purpose do patients in a satisfactory condition go to medical institutions? (1,2,3)

1. For a scheduled examination by a specialist; 2. For a planned procedure or operation; 3. For dispensary observation.

4 What conditions of the patient are one of the indications for hospitalization in a hospital? (1,2)

1. Moderate; 2. Heavy

5 What level of consciousness can a patient have in a state of moderate severity? (1,2,3)

1. Clear 2. Stupor 3. Sopor

2. Assessment of the patient's consciousness

1 Name the two main types of consciousness

1. clear; 2. Violated

2 List the three degrees of oppressed impairment of consciousness

1. Stupor; 2. Sopor (hibernation); 3. Coma.

3 Define stupor

state of stunnedness, from which the patient comes out for a short time during a conversation with him. Responds slowly when speaking

4 Define sopor

A condition in which the patient does not respond to others, accompanied by the preservation of pain sensitivity. When speaking, he answers in monosyllables (“yes-no”)

5 Define coma

A state in which the patient does not answer questions, does not respond to speech addressed to him. On examination, there is a disappearance of the main reflexes (including pain reflexes).

3. Position of the patient's body

1 List the types of body position of the patient. (1,2,3)

1. Active; 2. Passive; 3. Forced.

2 What is the patient's active position?

The ability of the patient to move independently, although the patient may experience various painful sensations

3 What is passive body position?

The condition of the patient in which he cannot independently change the position given to him.

4 Acceptance by the patient of a forced position allows? (1,2,3)

1. Make him feel better; 2. Improve the functioning of an organ or system (for example, drainage); 3. Correctly diagnose

5 What forced position do patients with inflammation of the pleura take? (1,2)

1. Lying position; 2. on the sore side.

4. Body type of the patient

1 List the body types of a person. (1,2,3)

1. Normosthenic; 2. Asthenic; 3. Hypersthenic

2 What costal angle is typical for each body type? (1,2,3)

1. Direct - for normosthenic; 2. Dumb - for hypersthenic; 3. Acute - for asthenic.

3 What are the dimensions of the chest? (1,2)

1. Anterior-posterior 2. Transverse

4 Describe the asthenic chest. (1,2,3,4,5)

1. Acute epigastric angle; 2. Wide intercostal spaces; 3. Oblique position of the ribs; 4. The shoulder blades do not fit snugly against the chest; 5. The transverse size of the chest significantly exceeds the anteroposterior

5 Describe the hypersthenic chest. (1,2,3,4,5)

1. Obtuse epigastric angle; 2. Narrow intercostal spaces; 3. More horizontal position of the ribs; 4. The shoulder blades are tightly pressed to the chest; 5. The anteroposterior size of the chest is approaching the transverse.

5. Skin examination

1 List the skin color options. (1,2,3,4,5,6).

1. Normal (pale pink for the European race); 2. Paleness; 3. Hyperemia; 4. Cyanosis; 5. Jaundice; 6. Bronze (brown) staining.

2 List the types of jaundice by pathogenesis? (1,2,3)

1. suprahepatic (hemolytic); 2. Hepatic (parenchymal); 3. Subhepatic (mechanical, or obstructive).

3 Define the term "bedsore"

These are deep ulcerations in severely debilitated patients with acute or chronic circulatory disorders and trophic disorders in places in areas of the skin subjected to constant pressure.

4 List the most commonly used places for determining edema by palpation. (1,2,3,4,5,6)

1. Shins; 2. Ankles; 3. sacrum; 4. Sternum; 5. Cheekbone; 6. Forehead.

5 List the reasons for the decrease in skin elasticity. (1,2)

1. Elderly and senile age 2. Dehydration of the body

6. Collection of complaints

1 Define symptom.

This is any sign of illness felt by the patient or determined by the doctor as a result of the examination of the patient.

2 Define the concept of syndrome

Consistently observed combination of symptoms united by a single pathogenesis (development mechanism)

3 List the main types of patient complaints. (1,2,3)

1. Basic 2. Additional; 3. Obtained by questioning the systems;

4 List the main characteristics of pain. (1,2,3,4,5,6,7)

1. Localization; 2. Character; 3. Irradiation; 4. Intensity; 5. Rhythm and duration; 6. Conditions of occurrence; 7. Terms of termination.

5 List the degrees of intensity of sensations. (1,2,3)

1. Weak 2. Moderate 3. Severe

7. Taking anamnesis of the present disease (anamnesis morbi)

1 What is an "anamnesis"? (1,2,3)

1. Information about the medical history; 2. About the diseases suffered by the patient; 3. About the patient's life history

2 List the factors contributing to the occurrence of this disease. (1,2,3,4,5,6)

1. Nutrition; 2. Heredity; 3. Lifestyle; 4. Physical activity; 5. Living conditions; 6. Working conditions.

3 What is described in the History of Present Illness section?

The occurrence, course and development of a real disease from its first manifestations to the moment of examination by the curator is described in detail in chronological order.

4 List the factors that reflect the dynamics of the present disease. (1,2,3,4)

1. Dynamics of present symptoms; 2. The appearance of new symptoms; 3. Frequency of exacerbations; 4. The occurrence of complications of the disease

5 Specify the methods of influence on the course of the present disease. (1,2,3,4)

1. Drug treatment; 2. Physiotherapy; 3. Sanatorium treatment; 4. Compliance with the regime of work and rest.

8. Collecting an anamnesis of life (Anamnesis vitae)

1 List bad habits that affect the development and prognosis of diseases? (1,2,3,4)

1. Smoking; 2. Drinking alcohol; 3. Drug use; 4. Substance abuse.

2 What factors of work history influence the development and course of diseases. (1,2,3,4,5,6)

1. Night shifts; 2. Hypothermia; 3. Long-term static stress; 4. Physical and emotional overstrain; 5. Temperature regime; 6. Exposure to toxic, chemical compounds, dust

3 Excessive consumption of what foods affects the prognosis of the course of cardiovascular diseases? (1,2)

1. Liquid; 2. Salt

4 What does an epidemiological history include? (1,2,3,4,5)

1. Contacts with infectious patients; 2. Stay in an endemic focus; 3. Bites of insects and animals; 4. Injuries; 5. Medical and cosmetic procedures

5 Immediate relatives whose presence of diseases is an important risk factor for the development of diseases in a patient include: (1,2,3)

1. Parents; 2. Brothers and sisters; 3. Children.

9. Anthropometry

1 What values are used to estimate body mass index? (1,2)

1. Growth; 2. Weights.

2 In accordance with WHO recommendations, indicate the normal values of BMI

18.50 - 24.99

3 What formulas allow for the assessment of anthropological indicators? (1,2,3)

1. Growth index; 2. Weight and height index; 3. Coefficient of proportionality.

4 Specify the anthropometric indicators that are obtained during the examination of the patient. (1,2,3,4)

1. Standing body length; 2. Sitting body length; 3. Weight; 4. Chest circumference.

5 What are the BMI values for underweight patients?

16.00 - 17.99 kg/m²

10. Diagnosis of abdominal obesity.

1 What nutritional factors are of great importance in the development of abdominal obesity? (1,2,3)

1. Meals mainly in the afternoon; 2. Excess consumption of easily digestible carbohydrates; 3. Excess consumption of animal fats.

2 List the main causes of abdominal obesity. (1,2)

1. Eating an excessive amount of calories; 2. Calorie intake/expenditure imbalance

3 What should be assessed when examining a patient with abdominal obesity? (1,2,3)

1. body weight; 2. Waist circumference; 3. Hip circumference

4 Name the main types of abdominal obesity. (1,2,3)

1. Primary; 2. Alimentary; 3. Secondary

5 Specify the main symptoms of hypothalamic abdominal obesity. (1,2)

1. Increased hunger; 2. Dulling the feeling of satiety with food

11. Thermometry

- 1 List the methods of thermometry. (1,2)**
1. Contact; 2. Contactless
- 2 Indicate what determines the temperature of the human body. (1,2)**
1. From the place of measurement; 2. From the time of day.
- 3 Indicate the types of body temperature depending on the thermometry data. (1,2,3,4,5,6)**
1. Hypothermia; 2. Norm; 3. Subfebrile; 4. Febrile; 5. Pyretic; 6. Hyperpyretic.
- 4 Specify the places for measuring body temperature. (1,2,3,4)**
1. Armpit; 2. Oral cavity; 3. Rectum; 4. Vagina.
- 5 Specify the main signs of fever. (1,2,3)**
1. Presence of infection; 2. Temperature rise above 37.00s; 3. The presence of pyrogens in the body

12. Types of temperature curves

- 1 What is a laxative fever? (1,2)**
1. Daily temperature fluctuations exceeding 10s; 2. Prolonged increase in body temperature.
- 2 What is a persistent fever characterized by? (1,2)**
1. Temperature fluctuations not exceeding 10s; 2. Prolonged increase in body temperature.
- 3 Specify the signs of exhausting fever. (1,2,3,4)**
1. Evening increase in body temperature up to 39-410C; 2. Normal body temperature in the morning; 3. An increase in temperature is accompanied by chills; 4. The decrease in temperature is accompanied by sweating.
- 4 What characterizes undulating fever is (1,2):**
1. Gradual smooth increase in body temperature; 2. Gradual gradual decrease in body temperature.
- 5 What types of fever are typical for purulent and septic processes in the body? (1,2)**
1. Remitting (laxative); 2. Hectic (depleting).

13. Heart rate measurement

- 1 What indicators are evaluated during the study of arterial pulse? (1,2,3,4,5,6)**
1. Frequencies; 2. Rhythm; 3. Voltage; 4. Filling; 5. Shapes; 6. Quantities.

2 Describe how pulsus differens is seen on examination. (1,2)

1. Unequal filling of the pulse in symmetrical areas of the same limbs; 2. Unequal pulse value in symmetrical areas of the same limbs

3 How to identify a pulse deficit pulsus deficiens. (1,2)

With a pulse deficit, the number of pulse waves per unit of time is less than the number of heartbeats

4 Define tachycardia

Pulse more than 80 beats per minute

5 Define bradycardia

Pulse less than 60 beats per minute

14. Blood pressure measurement, orthostatic test, tilt test

1 Pulse arterial pressure is determined.

difference between systolic and diastolic blood pressure

2 What determines the value of blood pressure? (1,2,3)

1. The volume of cardiac output; 2. Elasticity of the vascular wall; 3. Total peripheral vascular resistance

3 What are the types of blood pressure? (1,2,3)

1. Systolic; 2. Diastolic; 3. Pulse.

4 Under what pathological conditions does a decrease in blood pressure occur. (1,2,3,4,5,6)

1. Hypotonic disease; 2. Adrenal insufficiency; 3. Vascular insufficiency (septic, anaphylactic shock, severe blood loss, dehydration, etc.); 4. Acute heart failure; 5. Dysfunction of the parathyroid glands; 6. Taking certain medications (diuretics, alpha and beta blockers, some types of antidepressants, etc.)

5 Specify the clinical signs of acute vascular insufficiency. (1,2,3,4,5,6,7,8)

1. Sharp weakness; 2. Noise in the ears; 3. Darkening in the eyes; 4. Shortness of breath; 5. Sharp pallor of the skin; 6. Cold, clammy sweat; 7. Decreased systolic and diastolic pressure; 8. Thready pulse.

15. Respiratory rate measurement

1 What is called shortness of breath? (1,2,3)

1. Change in the frequency of respiratory movements; 2. Change in the depth of respiratory movements; 3. Violation of the rhythm of breathing

2 How is the respiratory rate determined? (NPV)

The number of inhalations and exhalations in 1 minute is counted, observing the excursion of the chest.

3 Specify the signs of Kussmaul's pathological respiration. (1,2,3)

1. Rare breathing; 2. Deep breathing; 3. Noisy breathing

4 Specify the signs of Biot's pathological respiration. (1,2,3)

1. Periodic breathing; 2. Pauses appear; 3. Correct alternation of respiratory movements

5 Describe Cheyne-Stokes breathing

This is a progressive increase and then a decrease in the frequency and depth of breathing, which is replaced by periods of apnea.

16. Pulse oximetry

1 Define pulse oximetry

Method for measuring the percentage of oxyhemoglobin in arterial blood

2 What is the importance of pulse oximetry in clinical practice? (1,2)

1. Diagnosis of arterial hypoxemia; 2. Diagnosis of respiratory failure

3 Indicate the value of saturation in case of respiratory failure of the 1st degree

90-94%

4 Indicate the value of saturation in respiratory failure of the 2nd degree

75-89%

5 Indicate the value of saturation in case of respiratory failure of the 3rd degree

Less than 75%

17. Topographic lines of the chest and topographic areas of the abdomen.

1 Write the name of all topographic lines on the chest along the front surface. (1,2,3,4,5,6,7)

1. Anterior median line (linea mediana anterior); 2. Left sternal line (linea sternalis sinistra); 3. Right sternal line (linea sternalis dextra); 4. Left parasternal (linea parasternalis sinistra); 5. Right parasternal (linea parasternalis dextra); 6. Left midclavicular line (linea medioclavicularis sinistra); 7. Right midclavicular line (linea medioclavicularis dextra).

2 Write the name of all topographic lines on the chest along the lateral surface. (1,2,3)

1. Anterior axillary line (linea axillaris anterior); 2. Middle axillary line (linea axillaris media); 3. Posterior axillary line (linea axillaris posterior).

3 Write the name of all topographic lines on the chest along the back surface. (1,2,3,4,5,6,7)

1. Left scapular line (linea scapularis sinistra); 2. Right scapular line (linea scapularis dextra); 3. Left paravertebral line (linea paravertebralis sinistra); 4. Right paravertebral line (linea paravertebralis dextra); 5. Left vertebral line (linea vertebralis sinistra); 6. Right vertebral line (linea vertebralis dextra); 7. Posterior median line (linea mediana posterior).

4 List the areas of the upper and middle abdomen. (1,2,3,4,5,6)

1. Left hypochondrium; 2. Epigastric; 3. Right hypochondrium; 4. Left side; 5. umbilical; 6. Right side.

5 List the areas of the lower abdomen. (1,2,3)

1. Left iliac (inguinal); 2. Suprapubic; 4. Right iliac (inguinal).

18. Palpation l / y

1 How is the palpation of the lymph nodes performed?

Circular motions with fingertips

2 What groups of lymph nodes differ in research technique in comparison with other groups? (1,2)

1. Submandibular; 2. Axillary

3 List the basic principles of palpation of the lymph nodes. (1,2)

1. Top down; 2. Front to back.

4 List the signs of pathological changes in the lymph nodes. (1,2,3)

1. Dense consistency; 2. Adhered to each other and / or with the surrounding tissue; 3. Painful; 4. Skin change over l / y.

5 Describe the skin changes in the area of the lymph node, indicating its pathology. (1,2)

1. Hyperemia in the projection of the lymph node 2. Swelling in the projection of the lymph node

1. Hyperemia in the projection of the lymph node 2. Swelling in the projection of the lymph node

19. Palpation of the musculoskeletal system

1 What technique is used to assess muscle tone?

According to the degree of resistance during passive movements of the limbs

2 What parameters are evaluated in the description of the musculoskeletal system. (1,2,3,4)

1. Consistency, compaction of muscle areas; 2. Soreness on palpation; 3. Muscle tone of the upper and lower extremities; 4. Symmetry of muscle strength indicators.

3 What changes in the spine are physiological? (1,2)

1. Lordosis in the cervical and lumbar spine; 2. Kyphosis in the thoracic and sacral spine

4 What do you pay attention to when evaluating a skeleton? (1,2,3,4)

1. On the shape of the bones; 2. For the presence of deformations; 3. For pain on palpation and effleurage; 4. For the presence of pathological fractures

5 What causes the symptom of ossalgia?

Periosteum lesion

20. Determination of edema

1 Specify the factors of edema appearance. (1,2,3,4)

1. Hydrostatic factor; 2. Oncotic factor; 3. Permeability of the capillary wall in places of edema; 4. Hormonal mechanisms of pressure regulation

2 What is characteristic of "cardiac" edema? (1,2,3,4,5)

1. Pronounced peripheral acrocyanosis; 2. Cold skin in places of edema; 3. Dense in consistency; 4. Initial appearance on the feet, shins; 5. The appearance or intensification of edema in the evening.

3 In what places is edema determined by palpation? (1,2,3,4,5,6,7)

1. On the feet, 2. Legs; 3. In the area of the shoulder blades, 4. On the sacrum 5. On the sternum; 6. On the face (zygomatic area, forehead); 7. On the back of the hands

4 What is characteristic of "renal edema"? (1,2,3,4,5)

1. Pale skin; 2. Warm skin in places of edema; 3. Loose in consistency; 4. Initial appearance on the face, eyelids; 5. Appear in the morning, decrease in the evening, may disappear

5 What is typical for edema caused by inflammation? (1,2,3,4)

1. Soft in texture; 2. Hot; 3. Sharply painful; 4. Often asymmetrical

21. Percussion. Methodology

1 Percussion sound has the following characteristics (1,2,3,4):

1. Volume; 2. Timbre; 3. Frequency response; 4. Duration

2 **In what joint is the main movement carried out at the moment of striking with percussion "finger on finger"?**

wrist

3 **To what depth does a loud percussion sound penetrate deep into the tissue?**

7-8 cm

4 **How is the plessimeter finger positioned during topographic percussion?**

Parallel to the border of expected stupidity

5 **List the rules by which topographic percussion of the chest organs is performed. (1,2)**

1. Percussion is carried out along the ribs and intercostal spaces; 2. Percussion should be quiet.

22. Percussion of the abdomen

1 **What percussion sound is normally determined over the entire surface of the abdomen?**

tympanic

2 **What tasks does percussion of the abdomen solve? (1,2)**

1. Identification of free, encysted fluid in the abdominal cavity; 2. Determination of the boundaries and sizes of some organs of the abdominal cavity.

3 **Describe the algorithm for determining free fluid in the abdominal cavity in the horizontal position of the patient? (1,2,3)**

1. Percussion is carried out from the navel in the lateral direction until dullness, 2. Without removing the finger of the plessimeter, the patient turns to the opposite side; 3. Percussion continues with an assessment of the change in the nature of the sound

4 **How is the "wave symptom" determined to detect free fluid?**

With the control imposition of the assistant's palm on the middle of the abdomen, light percussion blows are applied from one side of the abdominal cavity. On the opposite side, palpation sensations of the "transfer wave" are determined.

5 **Describe the algorithm for determining free fluid in the abdominal cavity in the vertical position of the patient (1,2,3):**

1. Percussion is performed from the xiphoid process to the pubic joint in a standing position until dullness 2. Without removing the finger of the plessimeter, the patient moves to a horizontal position; 3. Percussion continues with an assessment of the change in the nature of the sound.

23. Auscultation. Methodology

1 List the optimal conditions for auscultation. (1,2,3)

1. The room should be quiet; 2. The room should be warm; 3. The stethoscope/phonendoscope is firmly pressed against the auscultation point on the patient's body

2 What causes vascular murmurs on auscultation?

With a violation of laminar blood flow in the vessels.

3 What sounds are best heard with a stethoscope?

Low frequency sounds

4 What sounds are best heard with a stethoscope?

high frequency

5 List the differences between a phonendoscope and a stethoscope. (1,2)

1. The phonendoscope has a membrane; 2. A phonendoscope can amplify sound

24. Auscultation of the abdomen

1 What is normally heard over the entire surface of the abdomen?

Moderately frequent live peristaltic noises.

2 List the tasks of auscultation of the abdomen (1,2,3)

1. Definition of peristalsis; 2. The presence of vascular noise; 3. Determination of the friction noise of the peritoneum

3 How many sections of the abdomen are listened to according to the rules for conducting auscultation?

9

4 In what part of the abdomen is the rub of peritoneal friction most often heard? (1,2)

1. In the right hypochondrium; 2. In the left hypochondrium

5 List the vessels that can be heard during auscultation of the anterior abdominal wall (1,2,3,4):

1. Abdominal aorta; 2. Iliac arteries; 3. Renal arteries; 4. Femoral arteries

25. Electrocardiograph device

1 What does an electrocardiograph record?

Potential difference (voltage) between different parts of the patient's body surface (or electrodes that combine potentials from several parts of the body).

2 What electrode is conditionally called "positive" ("active")?

The one of the two electrodes, in the presence of which there is a more positive potential, the readings of the electrocardiograph are shifted in the positive direction (the ECG line is shifted up).

3 What ECG recording parameters can be configured in the electrocardiograph? (1,2,3)

1. Gain (by amplitude); 2. Recording speed (horizontally); 3. Noise/artifact filters

4 What is used as the "negative" ("inactive") electrode when registering the aVR lead?

Conductor connecting yellow and green standard electrodes.

5 What is used as a "negative" ("inactive") electrode when registering chest leads according to Wilson?

Conductor connecting red, yellow and green standard electrodes

26. ECG recording technique

1 List the conditions for conducting a planned electrocardiographic study. (1,2)

1. ECG is recorded in a special room, remote from possible sources of electrical interference; 2. The couch should be located at a distance of at least 1.5–2 meters from the mains wires.

2 What conditions regarding the patient must be met before a planned ECG recording? (1,2)

1. The study is carried out after 10-15 minutes of rest; 2. And not earlier than 2 hours after eating.

3 List the steps to take before starting an ECG recording. (1,2,3)

1. Invite the patient to undress to the waist, expose the ankles; 2. Invite the patient to lie on the couch on his back; 3. Treat the electrode sites with a composition that degreases the skin and improves electrical conductivity (solution, spray, gel if necessary).

4 List the steps required during an ECG recording. (1,2,3)

1. Select the parameters of amplification and recording speed; 2. Register 12 leads in manual or automatic mode; 3. Make sure the quality of the ECG recording is satisfactory.

5 List the actions required after recording an ECG. (1,2)

1. Disconnect the electrodes from the patient, if necessary, remove the remains of the conductive composition (gel, paste); 2. Ensure that the ECG recording contains mandatory information (name, gender and age of the patient, date and time of registration).

27. Technique of applying ECG electrodes

- 1 **Specify the location of the red electrode for routine ECG recording.**
Right hand
- 2 **Specify the location of the yellow electrode for routine ECG recording**
Left hand
- 3 **Specify the location of the green electrode for routine ECG recording**
Left leg
- 4 **Specify the location of the black electrode for routine ECG recording**
Right leg
- 5 **Specify the electrodes used for registration of the second standard lead (according to Einthoven). (1,2)**
1. red (on the right hand); 2. Green (on the left leg)

28. The main problems that arise during the registration of ECG, and their solution

- 1 **What problem can arise when recording an ECG (1) if the room is not warm enough, and what is the way to solve it? (2)**
1. The patient's cold tremor may cause registration artifacts; 2. Raise the room temperature.
What ECG recording problem can be indicated by the presence of a high-frequency wave in all or most ECG leads (1) and what is the way to solve this problem? (2,3)
1. "Tip" from the AC mains; 2. You should enable the option "suppression of high-frequency artifacts"; 3. Disconnect the ECG machine from the mains and then switch to an autonomous power source).
- 2
- 3 **What ECG recording problem can be indicated by the presence of recording artifacts in one or more ECG leads?**
Loose fit of the corresponding electrode
- 4 **What are the reasons for the loose fit of the electrode during ECG recording? (1,2)**
1. Incorrect application technique: insufficient degreasing of the skin (for disposable adhesive electrodes); 2. Significant hairline.
- 5 **What are the possibilities for improving ECG registration in the presence of significant hairline? (1,2)**
1. Shaving hair; 2. Apply enough ECG gel

29. Long-term Holter ECG monitoring (HM-ECG)

1 What preliminary preparation is necessary for a patient who is scheduled for long-term Holter ECG monitoring? (1,2)

1. Hygienic treatment (shower, bath); 2. If necessary, shaving the hair in the places where the electrodes are applied

2 What are the absolute contraindications for long-term Holter ECG monitoring? (1)

Violations of the patient's behavior (high risk of damage to the device or dismantling of the electrodes).

3 What are the relative contraindications for long-term Holter ECG monitoring? (1)

Violation of the integrity of the skin, increased sensitivity of the skin to the adhesive and conductive compounds used in the electrode

4 What should not be done to the patient during long-term Holter ECG monitoring? (1,2,3)

1. Wash in the shower or bath (there are models that allow you to take a shower subject to certain conditions); 2. Conduct MRI; 3. Perform significant physical activity that may damage the device or remove the electrodes

5 What should not be done (down to the minimum possible) for a patient during long-term Holter ECG monitoring? (1,2,3)

1. Be close to sources of electromagnetic radiation (mobile phones during a conversation or exchange of digital data, microwave ovens, power lines and transformers, strong magnets); 2. Conduct X-ray studies (X-ray, CT scan); 3. Use alcohol and other psychoactive substances.

30. Ambulatory blood pressure monitoring (ABPM)

1 What are the absolute contraindications for ambulatory blood pressure monitoring? (1,2,3)

1. Violations of the patient's behavior (high risk of damage to the device or violation of the rules for measuring blood pressure); 2. Damage to the skin or injuries of the upper limbs; 3. Diseases of the blood system during the period of exacerbation and / or with impaired blood clotting.

2 What are the relative contraindications for ambulatory blood pressure monitoring? (1,2,3)

1. Complaints of the patient about the poor tolerability of the study; 2. Severe heart rhythm and conduction disturbances; 3. The level of systolic blood pressure is more than 200 mm Hg. Art.

3 Which arm is recommended for a ABPM cuff?

On the "non-working" hand (left for right-handers, right for left-handers).

4 How is it recommended that the patient behave during automatic BP measurement as part of ABPM? (1,2,3)

1. If possible, stop or not move; 2. If possible, unbend the arm with the cuff, relax and lower it down; 3. Do not cross your legs or tense your leg muscles.

5 What should a patient record in a patient diary during ABPM? (1,2,3)

1. Exact time of onset and description of symptoms (if any); 2. Time of sleep and wakefulness; 3. Time and degree of physical activity

31. Study of the function of external respiration. Bodyplethysmography.

1 What is spirometry (examination of respiratory function)?

Measurement of tidal volumes and capacities, as well as volumetric velocities of air movement during quiet breathing and performing various breathing maneuvers.

2 What conditions must be observed before spirometry (research FVD)? (1,2)

1. The study is conducted in the morning; 2. The study is carried out on an empty stomach or not earlier than 2 hours after a light breakfast.

3 What are the absolute contraindications for spirometry? (1,2,3)

1. Acute conditions in which the performance of forced respiratory maneuvers may pose a threat to the health and life of the patient; 2. Impossibility of productive contact with the patient (cognitive impairment, language barrier, young children); 3. Diseases and injuries of the maxillofacial region, which do not allow sealing the connection with the device tube

4 What procedure should be performed daily (or more often) to ensure the accuracy of spirometry data (RF study)?

instrument calibration

5 What is "body plethysmography"?

Examination of the function of external respiration in a special sealed chamber, which allows determining the intrathoracic volumes of gas, including the residual volume of the lungs (RLR).

32. Electromyography

1 What is "electromyography"?

This is the registration of electrical potentials arising in human skeletal muscles.

2 What are the contraindications for electromyography? (1,2,3,4)

1. Violation of the integrity of the skin at the site of study; 2. Severe cardiac arrhythmias or a pacemaker (when conducting EMG with stimulation); 3. Epilepsy or other paroxysmal conditions; 4. Significant deviations in mental status.

3 Specify the two main options for conducting electromyography. (1,2)

1. Stimulation (superficial); 2. Needle (local).

4 What preparation is required before electromyography (1,2,3)?

1. For 24 hours - the abolition of certain drugs (tranquilizers, muscle relaxants and other drugs that affect the nervous and muscular systems); 2. Abstinence from alcohol; 3. For 2-3 hours - do not drink strong tea and coffee, caffeinated drinks, chocolate.

5 List the main indications for electromyography (1,2,3,4,5,6)

1. Any violation of motor functions; 2. Delays in motor development that are not directly related to brain damage; 3. Muscle weakness; 4. Increased muscle fatigue; 5. Pain in the muscles, muscle spasms; 6. Violations of sensitivity in the limbs.

33. Complete blood count

1 List requirements for blood sampling for clinical analysis (1,2,3).

1. Blood sampling is carried out in the morning; 2. Not less than 4 hours from the last meal; 3. Blood sampling is carried out from the central vein or from the fingertip.

2 What does a clinical blood test include? (1,2,3,4,5,6)

1. Determination of hemoglobin concentration, 2. Numbers of red blood cells and their characteristics (MCV, MCH, MCHC), 3. Counting the total number of leukocytes, 4. Determination of the leukocyte formula (percentage of different leukocytes), 5. Numbers and characteristics of platelets, 6. Definition erythrocyte sedimentation rate (ESR).

3 What is the most common reason for an increase in ESR? (1,2)

1. An increase in the plasma content of coarse proteins (fibrinogen, alpha and gamma globulins, paraproteins); 2. Decreased albumin content

4 What condition is most often indicated by neutrophilic leukocytosis?

Bacterial inflammation

5 What does the expression "leukocyte formula shift to the left" mean?

An increase in the number of young and immature forms of neutrophils (stab, metamyelocytes, myelocytes, etc.), which is typical for a general inflammatory reaction in a bacterial infection.

34. General analysis of urine. Daily proteinuria.

1 What does a general clinical urinalysis include? (1,2,3)

1. Determination of the physical properties of urine; 2. Chemical research; 3. Microscopy of sediment.

2 What causes can lead to the appearance of urine of red shades up to bright red? (1,2,3)

1. Blood (erythrocytes) in the urine; 2. Pigments after eating beets, blueberries, carrots, etc.; 3. Metabolites of various drugs (analgin, aspirin, etc.).

3 What causes brown-colored urine, possibly foaming ("beer-colored urine")? (1,2)

1. An increase in the level of bilirubin in the urine, 2. Perhaps in combination with an increased level of bile acids.

4 How soon after urine sampling should sediment microscopy be performed?

Not later than 2 hours after urine collection

5 How to collect urine to assess daily proteinuria (1,2,3,4,5)?

1. All urine is collected during the day in a clean container with a volume of 2-3 liters; 2. In this case, in the morning after sleep, you need to urinate into the toilet; 3. Then all urine is collected within 24 hours (including the morning portion the next day!); 4. In the morning, the urine collected during the day is mixed and about 100 ml is taken into a special container for collecting urine; 5. The referral documentation also indicates the total volume of urine collected.

35. Biochemical analysis of blood, urine.

1 Describe the procedure for taking blood for biochemical analysis.

The patient comes for blood sampling in the morning, on an empty stomach; blood is taken from a central vein, preferably using a disposable vacuum system (vacutainer).

2 In which test tube should blood be taken for biochemical analysis? (1,2)

1. Blood is taken into a test tube without filler or with a coagulation activator applied to the inner surface (for example, silicon dioxide).
2. Tubes containing a gel to separate the clot and serum during centrifugation or tubes for accelerated analysis (clotting activator - thrombin) can also be used.

3 List the main indicators of a biochemical blood test, which can be used to evaluate protein metabolism in the body. (1,2,3)

1. Total protein; 2. Albumin; 3. Urea

4 List the main indicators of a biochemical blood test, which can be used to assess the state of the urinary system. (1,2,3,4)

1. Urea; 2. Creatinine; 3. Uric acid; 4. Potassium.

5 What causes the appearance of glucose in the urine? (1,2,3)

1. Diabetes mellitus with glycemia above 8–9 mmol/l; 2. Increased filtration of glucose in case of damage to the glomeruli; 3. Decreased glucose reabsorption due to tubular damage or the use of modern hypoglycemic drugs ("gliptins").

36. Urinalysis according to Nechiporenko. Urinalysis according to Zimnitsky.

1 For what purpose can a urine test be carried out using the Nechiporenko method?

For a more accurate determination of the quantitative content of erythrocytes, leukocytes and cylinders (per unit volume of urine, and not per field of view when examining sediment).

2 What is the purpose of urinalysis using the Zimnitsky method?

Evaluation of the functional (concentration and excretory) ability of the kidneys under normal conditions.

3 How is urine collected for analysis according to the method of S. S. Zimnitsky?

On the morning of the study, the patient empties the bladder into the toilet at 6:00 a.m.; then it is necessary to collect urine during the day, every three hours in a separate numbered container (6-9 hours, 9-12 hours, 12-15 hours, 15-18 hours, 18-21 hours, 21-24 (0) hours, 0-3 hours, 3-6 hours).

4 What indicators are examined in the analysis of urine according to Zimnitsky? (1,2)

1. Volume of urine; 2. Density of urine

What can indicate a decrease in the maximum density of urine (hypostenuria) and a decrease in the spread of urine densities during the day (isosthenuria) in the analysis of urine according to Zimnitsky?

5

This may be one of the signs of a decrease in the concentration function of the kidneys during their initial (before clinical manifestation) damage.

37. Coagulogram

1 In which tube should blood be taken for a coagulogram?

To study the coagulogram, blood is taken into a test tube with a coagulation inhibitor (citrate).

2 What are the main indicators included in the coagulogram (1,2,3,4,5)?

1. Prothrombin time (PT); 2. Prothrombin index (PTI); 3. International normalized ratio (INR); 4. Activated partial thromboplastin time (APTT); 5. Fibrinogen level.

3 What is INR (1) and how is it calculated? (2)

1. INR is an international normalized ratio; 2. The time it takes for a patient's blood sample to coagulate divided by the time it takes for a healthy person's blood to coagulate.

4 What change in the coagulation system is evidenced by an increase in INR? (1)

About the development of hypocoagulation

5 What is prothrombin time?

This is the time of formation of a fibrin clot in plasma when calcium chloride and thromboplastin are added to it.

38. Sowing blood, urine for sterility with the determination of sensitivity to antibiotics.

1 What is a "blood (urine) culture"?

This is a microbiological study in which blood (urine) samples of a patient are placed on a number of nutrient media for various microorganisms and cultured until colonies are obtained, which can then be used to determine the type of microorganism, and subsequently its sensitivity to various antibacterial agents.

2 When should blood be taken for microbiological testing?

Best of all - at the peak of the fever

3 How long does it take to conduct a microbiological study of blood (urine)? (1,2)

1. Microorganism culture needs 3 to 5 days; 2. To determine sensitivity to antibiotics - another 1 or 2 days.

4 How many times is it recommended to take blood for sterility in a patient with suspected sepsis?

three times

5 What blood (urine) culture result is considered "positive"?

At a concentration of microorganisms 10^5 CFU/ml and above.

39. Fecal analysis

1 What preparation is required before taking material for fecal analysis? (1,2)

1. Avoid taking drugs that affect the characteristics of feces (laxatives, antibiotics, suppositories, iron and bismuth preparations, etc.); 2. The day before, do not perform per rectum procedures (recto- or colonoscopy, enema, etc.).

2 How is material taken for a general analysis of feces (coprogram)? (1,2)

1. After preparation in the morning, defecation is performed into a clean vessel / pot, or into the toilet bowl on a sterile diaper; 2. The feces is carried out with a spatula from the middle of the stool, avoiding the ingress of urine, menstrual flow or water from the toilet.

3 What is examined in the general analysis of feces (coprogram)? (1,2,3,4)

1. Organoleptic properties (color, appearance, smell, texture, presence of mucus); 2. pH; 3. Specific chemical reactions (for blood, bilirubin, etc.); 4. Microscopic examination

4 What is the name of the symptom in the analysis of feces with a pronounced greasy sheen and the detection of a large amount of undigested fats on microscopy?

Steatorrhea

5 What is "creator"?

Presence of large amounts of undigested food in the stool

40. Sputum tests

1 What preparation is required before collecting material for a general sputum analysis?

Thoroughly brush your teeth the night before the exam

2 How is material collected for general sputum analysis?

In the morning after the toilet of the oral cavity (thorough rinsing with boiled water), but without brushing the teeth, the patient expectorates sputum into a special disposable container (at least 1 ml of sputum), after which the container is closed with a tight lid and, together with the referral, is transferred to the laboratory

3 How is material taken for microbiological analysis of sputum?

In the morning after the toilet of the oral cavity (thorough rinsing with boiled water), but without brushing the teeth, the patient expectorates sputum into a special sterile container (at least 1 ml of sputum), after which the container is closed with a tight lid and, together with the referral, is transferred to the laboratory

4 What parameters are determined in the general sputum analysis (1,2,3)?

1. General appearance and color of sputum, 2. Character of sputum (serous, mucous, mucopurulent, purulent, putrefactive, hemorrhagic), 3. Microscopic examination: the presence of cells (epithelium, alveolar macrophages, leukocytes, erythrocytes, atypical cells, etc.), microorganisms and specific elements (for example, Kurshman spirals, Charcot-Leyden crystals, elastic fibers, etc.).

5 What does the detection of elastic fibers in the sputum indicate?

About the collapse of the lung tissue (for example, with an abscess, bronchiectasis, etc.).

41. Radiography. Mammography

1 Define the term radiography.

This is a method of studying the internal structure of objects that are projected using X-rays onto a special film or paper.

2 List the types of x-rays. (1,2)

1. Survey 2. Sight

3 List the indications for radiography. (1,2)

1. Injuries and diseases of bones, joints, spine; 2. Diagnosis of diseases of the chest and abdominal cavity

4 Define "r"fluoroscopy".

This is an X-ray examination method in which an image of an object is obtained on a luminous (fluorescent) screen.

5 Define the term "mammography".

This is an x-ray method for examining the mammary glands.

42. Radiography with contrast.

1 What is rX-ray with contrast?

This is a group of methods of X-ray examination based on the contrasting of hollow anatomical formations with radiopaque preparations.

2 List the types of radiopaque agents. (1,2)

1. Positive (absorbing X-rays more than the surrounding tissues): containing iodine and barium; 2. Negative (slightly absorb x-rays): oxygen, air.

3 What does barium sulfate fluoroscopy allow to examine? (1,2,3)

1. Esophagus; 2. Stomach; 3. Intestine

4 For contrasting what is X-ray with iodine preparations used? (1,2,3)

1. Kidney; 2. Pelvic organs; 3. Vessels.

5 What is the most common complication associated with the use of radiopaque agents?

Allergic reaction

43. Computed tomography. CT with I/V bolus contrast.

1 Define the concept of "computed tomography"

This is a method of non-destructive layer-by-layer study of the internal structure of an organ or tissue, based on the measurement and complex computer processing of the difference in X-ray attenuation by tissues of different density.

2 Which structures are imaged using computed tomography? (1,2,3,4,5,6,7)

1. Bones; 2. Soft tissues; 3. Blood vessels; 4. Chest; 5. Brain; 6. Abdominal organs; 7. Pelvic organs.

3 What radiopaque preparations are used in computed tomography?

Iodine-containing contrast agents.

4 List contraindications for CT with contrast. (1,2,3,4,5,6)

1. Individual intolerance to the contrast agent; 2. Renal failure (decrease in GFR < 30 ml/min/1.73 m²); 3. Autonomous diseases; 4. Hyperthyroidism and other pathology of the thyroid gland; 5. Severe bronchial asthma 6. Pregnancy.

5 Define the term "to computed tomographic angiography»

This is a highly informative research method, which includes computed tomography with intravenous administration of a radiopaque substance in order to obtain images of the vascular system.

44. MRI. MRI angiography.

1 Magnetic resonance imaging is (1):

A method for obtaining tomographic medical images for studying internal organs and tissues using the phenomenon of nuclear magnetic resonance, based on measuring the electromagnetic response of atomic nuclei of hydrogen atoms located in a strong constant magnetic field in response to their excitation by a certain combination of electromagnetic waves.

2 What structures are examined using MRI (1,2,3,4,5):

1. Cartilage; 2. Muscles; 3. Ligaments; 4. Brain and spinal cord; 5. Vessels.

3 Contraindications for MRI (1,2,3):

1. The presence of electronic medical devices in the body (electronic pacemakers, electronic devices for dosing drugs, etc.). 2. The presence of metal elements in the area under study (prostheses, clips, fragments). 3. Discrepancy between the parameters of the device and the anthropometric data of the patient in the study using MRI systems with an insufficient size of the magnet gap and / or a rigid receiving coil.

4 What contrast agents are used when performing magnetic resonance imaging (1):

Gadolinium-containing preparations

5 Contraindications for MRI with contrast (1,2,3,4,5,6,7):

1. Individual intolerance to the contrast agent; 2. Hemolytic anemia; 3. Pregnancy at any time; 4. Renal failure (decrease in GFR < 60 ml/min/1.73 m²); 5. Severe liver failure; 6. Severe bronchial asthma; 7. Mental illness.

45. PET.

1 Define the concept of positron emission tomography.

Radionuclide tomographic method for studying internal organs. The method is based on the registration of a pair of gamma quanta arising from the annihilation of positrons with electrons. Positrons are produced by the positron beta decay of a radionuclide that is part of a radiopharmaceutical that is introduced into the body before research.

2 Define "radiopharmaceutical."

A substance consisting of a radionuclide and a transport molecule with a high binding capacity to a tissue or a specific function of a human organ

3 What is the most commonly used radioisotope in diagnostic nuclear medicine?

Technetium-99m

4 In what areas of medicine is positron emission tomography used? (1,2,3):

1. Oncology; 2. Cardiology; 3. Neurology.

5 What does the doctor evaluate when performing positron emission tomography? (1)

metabolic rate

46. Echocardiography

1 Define the term echocardiography.

This is an ultrasound method aimed at studying the morphological and functional changes in the heart and its valvular apparatus.

2 List the types of echocardiography. (1,2,3,4)

1. Transthoracic; 2. Transesophageal; 3. Intracardiac; 4. Stress echocardiography

3 List indications for stress echocardiography. (1,2,3)

1. Diagnosis of coronary artery disease; 2. Assessment of myocardial viability; 3. Assessment of the LV contractility reserve.

4 List contraindications for transesophageal echocardiography. (1,2)

1. Pathology of the esophagus (tumors, diverticula, fistulas, strictures, esophagitis); 2. Bleeding from the upper gastrointestinal tract.

5 List the modes of echocardiography. (1,2,3,4,5,6)

1. 2D (B-mode); 2. M-mode; 3. Constant wave doppler; 4. Pulsed wave doppler; 5. Color Doppler mapping; 6. Tissue doppler.

47. Ultrasound

1 Define the concept of "ultrasound dopplerography of vessels".

This is a technique for ultrasound examination of blood vessels, based on the use of the Doppler effect, able to demonstrate the direction and speed of blood flow in the vessels

2 What does a duplex vascular scan include? (1,2)

1. Visualization of blood vessels; 2. Blood flow analysis.

3 What does Doppler ultrasonography of blood vessels investigate? (1,2,3,4)

1. Condition of vessel walls; 2. The course of the vessels; 3. Vessel lumen; 4. Hemodynamic characteristics of vessels.

4 What is triplex vascular scanning? (1,2)

1. Study of the anatomy of blood vessels; 2. Assessment of blood flow; 3. Accurate assessment of vascular patency in color mode

5 How should a patient be prepared for an ultrasound scan?

No preparation required

48. Ultrasound of the abdominal organs, kidneys, pelvic organs.

1 Define the term "ultrasound".

This is a method of non-invasive examination of the human body using ultrasonic waves, based on the piezoelectric effect.

2 The study of what organs and structures includes ultrasound of the abdominal organs? (1,2,3,4,5)

1. Gallbladder with ducts; 2. Pancreas; 3. Liver; 4. Abdominal aorta; 5. Spleen.

3 What does preparation include for ultrasound? (1,2,3,4)

1. Diet with restriction of fatty and gas-forming food 3 days before the study; 2. The study is carried out on an empty stomach (the last meal and water 8 hours before the study); 3. Refrain from chewing gum and smoking before the study; 4. Accept the use of medicines approved by the doctor

4 What organs and structures are examined during an ultrasound of the kidneys? (1,2,3,4,5)

1. Kidneys; 2. Ureters; 3. Adrenal glands; 4. Bladder; 5. Renal arteries.

5 What accesses are used during ultrasound of the pelvic organs in women? (1,2,3)

1. Transabdominally; 2. Transvaginally; 3. Transrectal.

49. Esophagogastroduodenoscopy (EGDS).

1 Define EGDS

This is an endoscopic examination that allows you to examine the internal lumen and mucous membrane of the esophagus, stomach and duodenum

2 List the organs that can be examined during EGDS. (1,2,3)

1. Esophagus; 2. Stomach; 3. 12 duodenum

3 List the types of EGDS. (1,2,3,4,5)

1. Scheduled diagnostic (screening); 2. Planned control; 3. Medical planned; 4. Emergency diagnostic; 5. Medical emergency.

4 Describe the procedure for preparing a patient for EGDS. (1,2,3,4,5,6)

1. Cessation of food intake 7-8 hours before the start of the study; 2. Stopping fluid intake 3-4 hours before the start of the study; 3. Irrigate the pharynx and pharynx with an aerosol spray of 10% lidocaine solution; 4. Put the patient on the left side; 5. Put the diaper under the patient's head; 6. Insert a mouthpiece with a hole into the patient's mouth.

5 List the manipulations performed during diagnostic EGDS. (1,2,3)

1. Biopsy; 2. Endoscopic parietal pH-metry; 3. Chromogastrospecty; 3. Foreign body removal

50. Fibrocolonoscopy (FCS).

1 Define FCC.

This is an endoscopic examination that allows you to examine the internal lumen and the mucous membrane of all parts of the colon and rectum

2 List the sections of the intestine that can be examined with FCS. (1,2,3,4,5,6,7)

1. The caecum; 2. Terminal ileum; 3. Ascending colon; 4. Transverse colon; 5. Descending colon; 6. Sigmoid colon; 7. Rectum

3 List ways to cleanse the colon before FCS. (1,2)

1. With the help of enemas; 2. With the help of medications

4 List the types of diagnostic FCS. (1,2)

1. Diagnostic (screening); 2. Control.

5 List the manipulations performed during diagnostic FCS. (1,2)

1. Biopsy; 2. Foreign body removal

51. Sigmoidoscopy

1 Define sigmoidoscopy

The technique of endoscopic examination of the lower intestine at 25-30 cm from the anus.

2 What parts of the intestine can be examined with the help of sigmoidoscopy? (1,2)

1. Rectum; 2. Lower sigmoid colon

- 3 **What is the difference between a sigmoidoscope and a fibrocolonoscope? (1,2,3)**
1. Long (short about 35-40 cm); 2. Diameter (2 times thicker); 3. Rigidity (does not bend).
- 4 **In what position is the patient placed during sigmoidoscopy? (1.2) Which position is preferred? (3)**
1. Knee-elbow; 2. On the left side; 3. Knee-elbow position is preferred
- 5 **List the manipulations performed during diagnostic sigmoidoscopy. (1,2)**
1. Biopsy; 2. Foreign body removal

52. Bronchoscopy

- 1 **Define tracheobronchoscopy.**
This is a research method for direct examination and assessment of the condition of the mucous membranes of the tracheobronchial tree: the trachea and bronchi using a special device - a bronchoscope.
- 2 **List the types of tracheobronchoscopy (1,2)**
1. Fibrobronchoscopy; 2. Rigid bronchoscopy
- 3 **What route of introduction of a fiberoptic bronchoscope is used during tracheobronchoscopy under local anesthesia. (1,2)**
1. Transnasal; 2. Transoral.
When performing local anesthesia, before performing fibrobronchoscopy, at the first stage, (1) is processed, an anesthetic is used (2), and the concentration of anesthetic (3) is used. At the second stage, (4,5,6,7) is processed, an anesthetic (8) is used, and the concentration of anesthetic (9) is used.
- 4 **1. Zev of the larynx; 2. Lidocaine spray; 3. 10%; 4. Glottis; 5. Trachea; 6. Bifurcation of the trachea (karina); 7. Lobar bronchi; 8. Lidocaine solution; 9.2%.**
- 5 **List the manipulations performed during fibrobronchoscopy. (1,2,3)**
1. Biopsy; 2. Taking washing water; 3. The establishment of an endotracheal tube.

53. Cystoscopy

- 1 **Define urethrocystoscopy.**
This is an endoscopic examination that allows you to perform an examination and assessment of the condition of the urethra, bladder mucosa, orifices of the ureters.
- 2 **List the types of cystoscopes? (1,2)**

1. Rigid cystoscope; 2. Fibrocystoscope

3 What expansion media can be used for diagnostic cystoscopy? (1,2,3)

1. Sterile water; 2. Physiological solution; 3. A solution of 5% glucose.

4 What types of anesthesia are used during cystoscopy. (1,2,3,4)

1. Without anesthesia; 2. Local anesthesia with lidocaine; 3. Intravenous anesthesia; 4. Spinal anesthesia.

5 List the manipulations performed during diagnostic cystoscopy. (1,2,3,4)

1. Biopsy; 2. Chromocystoscopy; 3. Spectroscopy; 4. Retrograde ureteropyelography

54. Hysteroscopy

1 Define hysteroscopy

This is an endoscopic examination that allows you to perform an examination and assessment of the condition of the mucous membrane of the cervical canal, uterine cavity, orifices of the fallopian tubes.

2 List the types of hysteroscopes? (1,2,3)

1. Hard hysteroscope; 2. Office hysteroscope; 3. Fibrohysteroscope

3 What expansion media can be used for diagnostic cystoscopy? (1,2)

1. Saline solution; 2. 5% glucose solution

4 What types of anesthesia are used during cystoscopy. (1,2,3)

1. Without anesthesia (office hysteroscopy, fibrohysteroscopy); 2. Intravenous anesthesia; 3. Spinal anesthesia.

5 List the manipulations performed during diagnostic cystoscopy. (1,2,3)

1. Biopsy; 2. Spectroscopy; 3. Evaluation of the patency of the fallopian tubes.

55. Complications of endoscopic studies.

1 List the main complications of EGDS. (1,2,3)

1. Allergic reaction to the anesthetic; 2. Organ perforation; 3. Bleeding.

2 List the main complications of colonoscopy and sigmoidoscopy. (1,2)

1. Perforation of the intestine; 2. Bleeding.

3 List the main complications of bronchoscopy. (1,2,3,4,5)

1. Allergic reaction to the anesthetic; 2. Organ perforation; 3. Bleeding; 4. Bronchospasm; 5. Pneumonia.

4 List the main complications of cystoscopy. (1,2,3,4)

1. Infections in the cavity of the bladder; 2. The appearance of cystitis or erosion of the bladder; 3. Traumatic complications (ruptures of the urethra); 4. Perforation of the bladder wall

5 List the main complications of hysteroscopy. (1,2,3,4)

1. Perforation of the uterus; 2. Incomplete perforation of the uterus (false course); 3. Air embolism; 4. Inflammation of the uterus.

56. Prevention of complications of endoscopic examinations

1 In order to reduce the likelihood of complications during endoscopic examinations, it is necessary: (1,2,3,4,5,6)

1. Careful history taking; 2 Careful preparation of the patient for the study; 3. Compliance with the research methodology; 4. Reliable visualization of the studied areas; 5. Observation of patients in the hospital after removal of large neoplasms; 6. Termination of the study when the patient's condition worsens;

2 After endoscopic removal of large polyps, it is necessary: (1,2,3)

1. Observation of the patient in the hospital; 2. Control of hemoglobin and blood leukocytes; 3. Control endoscopic examination if a complication is suspected

3 If bleeding occurs after endoscopy, follow these steps: (1,2,3,4)

1. Assign bed rest; 2. Call the doctor on duty for examination; 3. Take a blood test for hemoglobin, hematocrit, red blood cell count; 4. Conduct a second endoscopic examination

4 If you suspect damage to the integrity of the wall of the organ under study, you must perform the following actions: (1,2,3)

1. Call the doctor on duty for examination; 2. Conduct a survey radiography of the abdominal organs; 3. Call a surgeon for examination to decide on the need for surgical treatment.

5 If an allergic reaction develops during an endoscopic examination, it is necessary: (1,2,3,4)

1. Stop the study; 2. Call a resuscitator for a consultation; 3. Intravenous administration of glucocorticoid hormones; 4. Hospitalize the patient in a hospital

57. Biopsy

1 Define the term "biopsy".

A research method in which cells or tissues (biopsy) are taken from the body in vivo for diagnostic or research purposes

2 List the methods of obtaining material for histological examination. (1,2,3,4,5,6,7)

1. Excisional biopsy; 2. Incisional biopsy; 3. Punch biopsy; 4. Trepan biopsy; 5. Core (core biopsy, core biopsy, cutting biopsy); 6. Scarifying (superficial) biopsy (shaving biopsy); 7. Loop biopsy.

3 List the methods of obtaining material for cytological examination. (1,2,3,4)

1. An imprint from a pathological formation (erosion, ulcer); 2. A smear-imprint from a pathological formation; 3. Fine needle aspiration biopsy (FNAB); 4. Aspiration biopsy

4 What is the fundamental difference between histological and cytological examination?

A cytological study differs from a histological one in that it does not examine tissue, but studies cells.

5 List the most informative types of histological examination

1. Histological examination 2. Immunohistochemistry (IHC) 3. Fluorescence in situ hybridization (FISH)

58. Laparocentesis

1 Define the term laparocentesis

Surgical procedure, which is a puncture of the abdominal wall to extract pathological contents

2 What instrument is used for laparocentesis? (1) What parts does it consist of? (2.3)

1. Trocar; 2. Stiletto; 3. Tube (tube, sleeve, cannula).

3 List the indications for laparocentesis. (1,2,3,4)

1. Evacuation of fluid from the abdominal cavity; 2. Establishing the nature of pathological exudate or transudate in the abdominal cavity in case of injuries and diseases; 3. Introduction of gas during abdominal radiography in case of suspected diaphragmatic rupture; 4. Introduction to the abdominal cavity of drugs

4 Describe the position of the patient during laparocentesis

The patient sits with his back supported on a chair, a container is placed on the floor between the legs to collect fluid.

5 At what point of the anterior abdominal wall is a puncture performed during laparocentesis?

In the middle of the distance from the navel to the pubis in the midline

59. Pleural puncture

1 Define the term "pleural puncture".

A surgical procedure that involves puncturing the chest wall and parietal pleura to extract fluid or gas

2 What instruments are used for pleural puncture? (1,2)

1. Trocar 2. Needle, caliber more than 1.0 mm

3 Where is a pleural puncture performed to remove fluid? (1,2,3)

1. In the VII or VIII intercostal space; 2. Along the posterior axillary or scapular line; 3. Along the upper edge of the underlying rib.

4 Where is a pleural puncture performed to remove gas? (1,2,3)

1. In II or III intercostal space; 2. Along the midclavicular line; 3. Strictly perpendicular to the upper edge of the underlying rib

5 Describe the position of the patient during a pleural puncture to remove fluid. (1,2)

1. Sitting on a chair, facing the back of the chair; 2. It is necessary to tilt the chest to the “healthy” side; 3. The patient should place the hand on the side of the puncture on the head or on the opposite shoulder.

60. Pericardial puncture

1 Define the term "pericardial puncture"

Diagnostic and therapeutic surgical manipulation, including puncture of the pericardial sac and evacuation of fluid (exudate, blood) from the pericardial cavity.

2 What instruments are used for pericardial puncture? (1,2)

1. Thin trocar; 2. Needle 15 cm long, caliber 1.2 - 1.5 mm.

3 List the two main methods of pericardial puncture.

1. Larrey method; 2. Marfan method;

4 Where is the pericardial puncture point according to the Larrey method?

The puncture point corresponds to the apex of the angle formed by the left costal arch (attachment of the cartilage of the VII rib to the sternum) and the base of the xiphoid process

5 Where is the pericardial puncture point according to the Marfan method?

The puncture point is located under the apex of the xiphoid process along the midline of the body.

61. Spinal (lumbar) puncture

1 Define the term "lumbar puncture"

Diagnostic surgical manipulation, which is the introduction of a needle into the subarachnoid space of the spinal cord at the lumbar level

2 Why is a lumbar puncture performed for diagnostic purposes? (1,2,3)

1. To measure CSF pressure; 2. Studies of the patency of the subarachnoid space of the spinal cord; 3. Determination of color, transparency and composition of cerebrospinal fluid.

3 Describe the positioning of the patient for a lumbar puncture in the supine position.

The patient is placed on a flat (hard) bed or on a high couch, on the right side. In order to avoid lateral arching of the spine, a pillow is placed under the lower back. The patient lies on his side, his head is tilted to the chest, the legs are bent at the knees, the hips are brought to the stomach, the stomach is pulled in, the back is arched.

4 Describe the position of the patient for lumbar puncture in the sitting position.

The patient is seated on a couch or across a narrow table with his back to its edge and dangling his legs. The patient's muscles should be relaxed, the back arched backwards, and the head tilted forward. To maintain balance, the patient rests his elbows on his hips or puts his hands on the back of a chair in front of him. An assistant holds the patient in this position.

5 Where is a lumbar puncture performed? (1,2)

1. Between III and IV lumbar vertebrae; 2. Between II and III lumbar vertebrae

62. Sternal puncture.

1 Define the term "sternal puncture".

Diagnostic surgical manipulation obtaining a bone marrow sample by puncture of the sternum

2 Where is the sternal puncture performed?

At the level of attachment of the III-IV ribs along the midline.

3 What instrument is used for sternal puncture?

Cashier needle

4 What types of preparations are prepared from bone marrow punctate? (1,2,3)

1. Thin smears from bone marrow fluid; 2. Smears of crushed lumps; 3. Histological sections by concentration, fixation and inclusion of lumps.

5 In which medical institutions can a sternal puncture be performed? (1,2)

1. In medical organizations that include a hospital; 2. In medical organizations that include a day hospital.

63. Main complications of invasive diagnostics

1 List the main complications of laparocentesis. (1,2,3,4)

1. Perforation of hollow organs; 2. Damage to the vessels of the greater omentum and the anterior abdominal wall, followed by bleeding; 3. Peritonitis; 4. Infection of the anterior abdominal wall.

2 List the main complications of pleural puncture. (1,2,3,4,5,6)

1. Trauma of intercostal arteries, nerves; 2. Secondary pneumo-, hemothorax; 3. Infectious complications; 4. Fainting, collapse, shock; 5. Injury to the subdiaphragmatic organs; 6. Bronchopleural fistula.

3 List the main complications of pericardial puncture. (1,2,3,4,5,6)

1. Pneumopericardium; 2. Pneumomediastinum; 3. Pneumothorax; 4. Perforation of the heart; 5. Arrhythmia; 6. Hypotension (when draining a large volume of effusion).

4 List the main complications of spinal (lumbar) puncture. (1,2,3,4,5,6,7)

1. Axial insertion; 2. Meningism; 3. Infectious complications; 4. Headaches; 5. Hemorrhagic complications; 6. Damage to roots; 7. Damage to the intervertebral disc with the formation of a disc herniation.

5 List the main complications of sternal puncture. (1,2)

1. Through puncture of the sternum; 2. Bleeding from the puncture site

64. Determination of blood group and Rh factor

1 Describe the process of determining blood groups of the AB0 system

Standard sera of the AB0 system of two different series of each group are applied to the plate under the appropriate designations so that two rows of three large drops (0.1 ml) are obtained in the following order from left to right: 0 (I), A (II), B (III). The test blood is applied one drop (0.01 ml) next to each drop of serum and the blood is mixed with serum. The progress of the reaction is monitored by gently rocking the plate for 5 minutes at room temperature.

2 Describe the process of testing for compatibility by blood groups of the AB0 system.

Two or three drops of the patient's blood serum are applied to the white plate, to which a ten times smaller drop of the donor's blood is added. The blood is mixed with the patient's serum, then the plate is periodically shaken for five minutes and the result of the reaction is simultaneously observed. The absence of agglutination of the donor's erythrocytes indicates the compatibility of the blood of the donor and the recipient in relation to ABO blood groups.

3 Describe the process for determining group membership when using zoliclones.

Apply Anti-A, Anti-B zoliclones to the tablet one large drop (0.1 ml) each under the appropriate inscriptions. Next to the drops of antibodies, apply with a glass rod one small drop of the test blood (0.01-0.03 ml). Separate glass rods mix the blood with the reagent. Within 3 minutes, observe the progress of the reaction with the zoliclones, while gently shaking the plate.

4 What methods can determine the Rh factor? (1,2)

1. Anti-rhesus serum (express method); 2. Zolicle anti-D.

5 What equipment is used in the determination of Rhesus affiliation with coliclones? (1,2,3,4,5,6,7)

1. Tsoliklon Anti-D; 2. Bottle with isotonic sodium chloride solution; 3. Tablets; 4. Glass rods; 5. Pipettes; 6. Magnifying glass; 7. Donor's transfusion medium container and/or patient's blood sample.